Register Number: 192425119

Question 1: Rainy Weather Accident Proportion

Research Question

Is the proportion of accidents during rainy weather significantly different from 30%?

Hypotheses

- H0: p = 0.30 (rainy-weather share of accidents equals 30%)
- H1: p ≠ 0.30 (two-sided test)
- Significance level α = 0.05

Formula

$z = (\hat{p} - p_o) / \sqrt{(p_o(1 - p_o) / n)}$

$\hat{p}$ = *number of rainy-weather accidents / total accidents*

Data Used (read from reference figure)

Weather condition	Accidents
Clear	42
Rainy	30
Foggy	15
Cloudy	13
Total	100

Calculation

- $\hat{p} = 30 / 100 = 0.30$
- $p_o = 0.30$, $n = 100$
- $z = (0.30 - 0.30) / \sqrt{(0.30 \times 0.70 / 100)} = 0.00$

Result

Quantity	Value
Observed proportion $\hat{p}$	0.30
z statistic	0.00
Two-sided p-value	1.00

Conclusion

The p-value (1.00) is far above $\alpha = 0.05$, so we fail to reject H0. The rainy-weather proportion of accidents does not differ significantly from 30%.

Graph

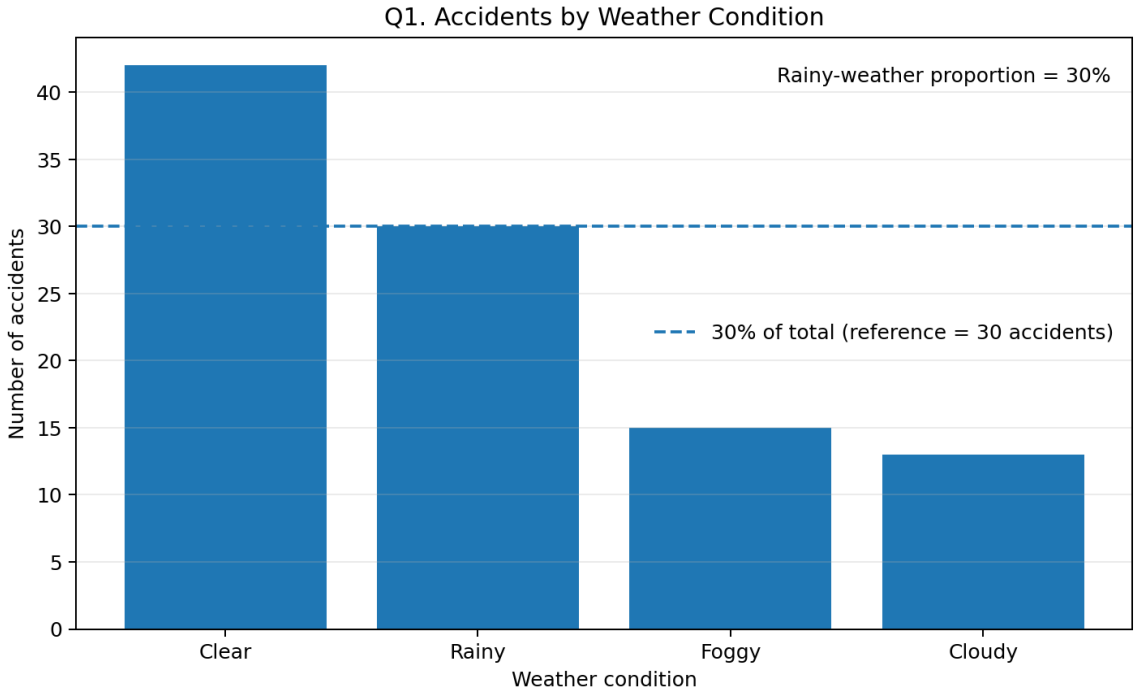


Figure 1. Reference graph for Question 1.

Note: the counts and values in this document were read from the reference figure. Re-run the test on the original project dataset in R (e.g. chisq.test, t.test, aov, cor.test) to confirm the final numbers before submission.